

Multiple discrete/continuous random variables

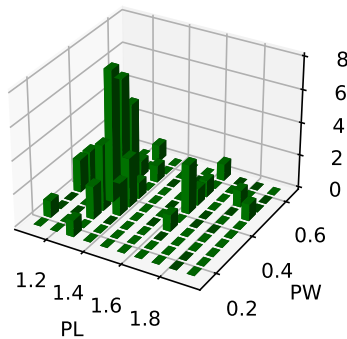
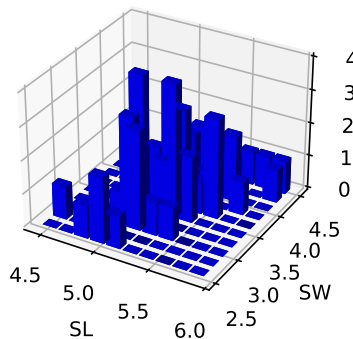
Andrew Thangaraj

IIT Madras

Subsection 3

Jointly continuous random variables

2D histograms: (SL, SW) and (PL, PW) for Class 0



- Count the number of (x, y) falling into a rectangular bin
- (SL, SW): Both continuous and they have a joint distribution
 - ▶ Same for (PL, PW)

Joint density in two dimensions

Definition (Joint density)

A function $f(x, y)$ is said to be a joint density function if

- $f(x, y) \geq 0$, i.e. f is non-negative
- $\iint_{-\infty}^{\infty} f(x, y) dx dy = 1$
- Technical: $f(x, y)$ is piecewise continuous in each variable

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 - Technical: $f(x, y)$ is piecewise continuous in each variable
-
- For every joint density $f(x, y)$, there exist two jointly distributed continuous random variables X and Y such that, for any two-dimensional region A ,

$$P((X, Y) \in A) = \iint_A f(x, y) dx dy$$

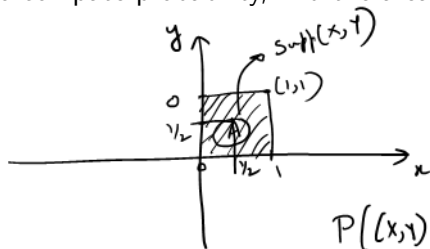
- $f(x, y)$, also denoted $f_{XY}(x, y)$, is called the joint density of X and Y
- $\text{supp}(X, Y) = \{(x, y) : f_{XY}(x, y) > 0\}$

Example: Uniform in the unit square

Let X and Y have joint density

$$f_{XY}(x, y) = \begin{cases} 1 & 0 < x < 1, 0 < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

- Picture the 3D plot of the joint density
- To compute probability, find the area of the region



$$\int_0^1 \int_0^1 1 \, dx \, dy = \int_0^1 \left[x \right]_0^1 dy = \int_0^1 1 \, dy = \left[y \right]_0^1 = 1$$

$$P((x, y) \in A) = \int_A \int 1 \, dx \, dy = \text{Area of } A$$

$$P(0 < x < 1/2, 0 < y < 1/2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

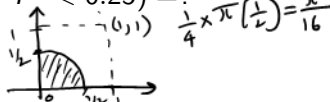
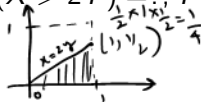
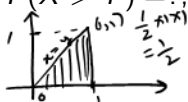
Example: Uniform in the unit square

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$$f_{XY}(x, y) = \begin{cases} 1 & 0 < x < 1, 0 < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$



- Picture the 3D plot of the joint density
- To compute probability, find the area of the region
- $P(0 < X < 0.1, 0 < Y < 0.1) = ?$, $P(0.5 < X < 0.6, 0 < Y < 0.1) = ?$, $P(0.9 < X < 1, 0.9 < Y < 1) = ?$
 $0.1 \times 0.1 = 0.01$ (grey), $0.1 \times 0.1 = 0.01$ (blue), $0.1 \times 0.1 = 0.01$ (green)
- $P(0 < X < 0.1) = ?$, $P(0.5 < Y < 0.6) = ?$
 $1 \times 0.1 = 0.1$ (grey), $1 \times 0.1 = 0.1$ (blue)
- $P(X > Y) = ?$, $P(X > 2Y) = ?$, $P(X^2 + Y^2 < 0.25) = ?$
 $\frac{1}{2} \times \pi \times (\frac{1}{2})^2 = \frac{\pi}{8}$ (grey), $\frac{1}{2} \times \pi \times (\frac{1}{2})^2 = \frac{\pi}{8}$ (blue), $\frac{1}{4} \times \pi \times (\frac{1}{2})^2 = \frac{\pi}{16}$ (grey)



2D uniform distribution

Fix some (reasonable) region D in $\mathbb{R}^2$ with total area $|D|$. We say that $(X, Y) \sim \text{Uniform}(D)$ if they have the joint density

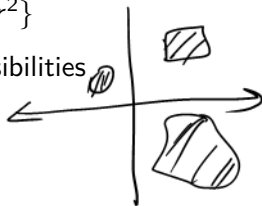
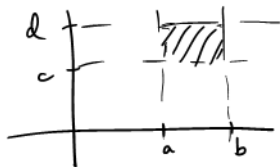
$$f_{XY}(x, y) = \begin{cases} \frac{1}{|D|} & (x, y) \in D, \\ 0 & \text{otherwise.} \end{cases}$$

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- Rectangle: $D = [a, b] \times [c, d] = \{(x, y) : a < x < b, c < y < d\}$
- Circle: $D = \{(x, y) : (x - x_0)^2 + (y - y_0)^2 \leq r^2\}$
- Multiple disjoint areas and so many other possibilities



2D uniform distribution

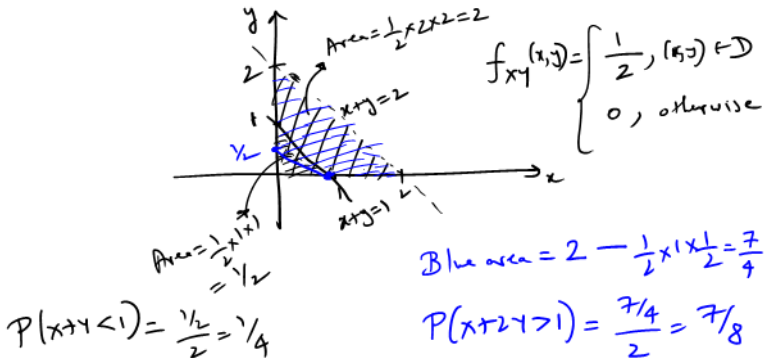
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- Circle: $D = \{(x, y) : (x - x_0)^2 + (y - y_0)^2 \leq r^2\}$
- Multiple disjoint areas and so many other possibilities
- For any sub-region A of D , $P((X, Y) \in A) = |A|/|D| = \frac{\text{Area}(A)}{\text{Area}(D)}$
- Uniform distribution is a good approximation for flat histograms

Problem: 2D uniform

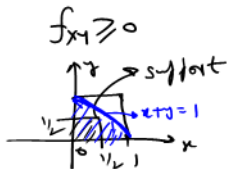
Let $(X, Y) \sim \text{Uniform}(D)$, where $D = \{(x, y) : x + y < 2, x > 0, y > 0\}$. Sketch the support and compute $P(X + Y < 1)$, $P(X + 2Y > 1)$.



Problem: 2D non-uniform

Let (X, Y) have joint density

$$f_{XY}(x, y) = \begin{cases} x + y, & 0 < x, y < 1, \\ 0, & \text{otherwise.} \end{cases}$$



Show that the above is a valid density. Find $P(X < 1/2 \text{ and } Y < 1/2)$, $P(X + Y < 1)$.

$$\int_{y=0}^1 \int_{x=0}^1 (x+y) dx dy = \int_{y=0}^1 \left(\frac{1}{2} + y \right) dy = \left. \frac{y}{2} + \frac{y^2}{2} \right|_0^1 = 1$$

$$\int_0^1 (x+y) dx = \left. \frac{x^2}{2} + yx \right|_0^1 = \frac{1}{2} + y$$

$$\begin{aligned} P(X < 1/2, Y < 1/2) &= \int_0^{1/2} \int_0^{1/2} (x+y) dx dy \\ &= \int_0^{1/2} \left(\frac{x^2}{2} + yx \right) \Big|_0^{1/2} dy \\ &= \int_0^{1/2} \left(\frac{1}{8} + \frac{y}{2} \right) dy = \left. \frac{y}{8} + \frac{y^2}{4} \right|_0^{1/2} \\ &= \frac{1}{8} \end{aligned}$$

$$\begin{aligned} P(X+Y < 1) &= \int_0^1 \int_0^{1-y} (x+y) dx dy = \int_0^1 \left(\frac{x^2}{2} + yx \right) \Big|_0^{1-y} dy \\ &= \int_0^1 \left(\frac{(1-y)^2}{2} + y(1-y) \right) dy = \int_0^1 \left(\frac{1}{2} - \frac{y}{2} \right) dy = \left. \frac{y}{2} - \frac{y^2}{6} \right|_0^1 = \frac{1}{2} - \frac{1}{6} = \frac{1}{3} \end{aligned}$$

Subsection 4

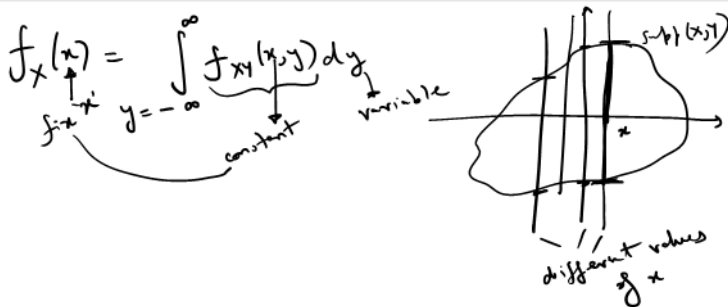
Marginal densities and independence

Marginal density

Theorem (Marginal density)

Suppose (X, Y) have joint density $f_{XY}(x, y)$. Then,

- X has the marginal density $f_X(x) = \int_{y=-\infty}^{\infty} f_{XY}(x, y) dy$
- Y has the marginal density $f_Y(y) = \int_{x=-\infty}^{\infty} f_{XY}(x, y) dx$



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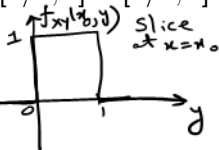
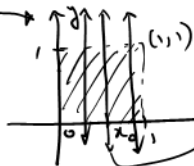
- The PDF of X and Y individually are called marginal densities
- The joint density exactly determines both the marginal densities

Examples: (Marginals do not determine joint)

- Uniform on unit square

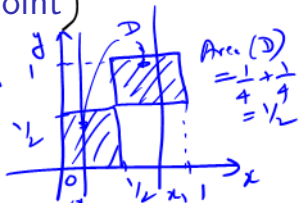
- $(X, Y) \sim \text{Uniform}(D)$, where

$$D = [0, 1/2] \times [0, 1/2] \cup [1/2, 1] \times [1/2, 1]$$

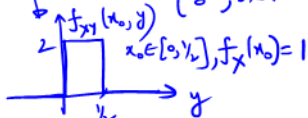


$$f_X(x_0) = \int_{y=0}^1 \underbrace{f_{XY}(x_0, y)}_1 dy = 1, \quad 0 < x_0 < 1$$

$X \sim \text{Uniform}[0, 1]$
 $Y \sim \text{Uniform}[0, 1]$



$$f_{XY}(x, y) = \begin{cases} 2, & (x, y) \in D \\ 0, & \text{otherwise} \end{cases}$$

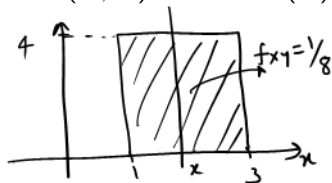


$$f_X(x) = 1, \quad 0 < x < 1$$

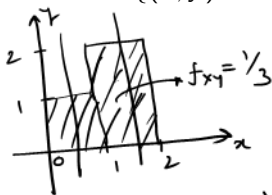
$X \sim \text{Uniform}[0, 1]$
 $Y \sim \text{Uniform}[0, 1]$

Examples: More uniform

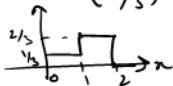
- $(X, Y) \sim \text{Uniform}(D)$, where $D = [1, 3] \times [0, 4]$
- $(X, Y) \sim \text{Uniform}(D)$, where $D = [0, 1] \times [0, 1] \cup [1, 2] \times [0, 2]$
- $(X, Y) \sim \text{Uniform}(D)$, where $D = \{(x, y) : x + y < 2, x > 0, y > 0\}$



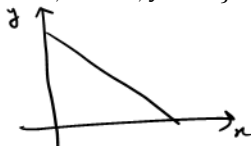
$$f_X(x) = \frac{1}{2}, 1 < x < 3$$
$$X \sim \text{Uniform}[1, 3]$$
$$Y \sim \text{Uniform}[0, 4]$$



$$f_X(x) = \begin{cases} \frac{1}{3}, & 0 < x < 1 \\ \frac{2}{3}, & 1 < x < 2 \end{cases}$$



$$f_Y(y) = \begin{cases} \frac{2}{3}, & 0 < y < 1 \\ \frac{1}{3}, & 1 < y < 2 \end{cases}$$



Problem

Consider the joint density

$$f_{XY}(x, y) = \begin{cases} x + y, & 0 < x, y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the marginals.

Independence

Theorem (Independence)

(X, Y) with joint density $f_{XY}(x, y)$ are independent if

$$f_{XY}(x, y) = f_X(x)f_Y(y),$$

where $f_X(x)$ and $f_Y(y)$ are the marginal densities.

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where $f_X(x)$ and $f_Y(y)$ are the marginal densities.

- Given the joint density, the marginals can be computed
- If the joint density is the product of the marginal densities, then X and Y are independent
- So, if independent, the marginals determine the joint density

Examples

- Uniform on unit square
- $(X, Y) \sim \text{Uniform}(D)$, where $D = [0, 1/2] \times [0, 1/2] \cup [1/2, 1] \times [1/2, 1]$
- $(X, Y) \sim \text{Uniform}(D)$, where $D = [1, 3] \times [0, 4]$
- $(X, Y) \sim \text{Uniform}(D)$, where $D = [0, 1] \times [0, 1] \cup [1, 2] \times [0, 2]$
- $(X, Y) \sim \text{Uniform}(D)$, where $D = \{(x, y) : x + y < 2, x > 0, y > 0\}$
-

$$f_{XY}(x, y) = \begin{cases} x + y, & 0 < x, y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Problem

Suppose $X \sim \text{Exp}(\lambda_1)$, $Y \sim \text{Exp}(\lambda_2)$ are independent random variables. Find their joint density and compute $P(X > Y)$.

$$f_X(x) = \lambda_1 e^{-\lambda_1 x}, x > 0$$

$$f_Y(y) = \lambda_2 e^{-\lambda_2 y}, y > 0$$

$$f_{X,Y}(x,y) = \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y}, x > 0, y > 0$$

$$\begin{aligned}
 P(X > Y) &= \int_{x=0}^{\infty} \left(\int_{y=0}^x \lambda_1 e^{-\lambda_1 x} \lambda_2 e^{-\lambda_2 y} dy \right) dx \\
 &= \int_{x=0}^{\infty} \lambda_1 e^{-\lambda_1 x} \left(-e^{-\lambda_2 y} \Big|_0^x \right) dx = \int_0^{\infty} \lambda_1 e^{-\lambda_1 x} (1 - e^{-\lambda_2 x}) dx \\
 &= \underbrace{\int_0^{\infty} \lambda_1 e^{-\lambda_1 x} dx}_1 - \int_0^{\infty} \lambda_1 e^{-(\lambda_1 + \lambda_2)x} dx \\
 &= 1 - \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{\lambda_2}{\lambda_1 + \lambda_2}
 \end{aligned}$$

Diagram illustrating the joint density function $f_{X,Y}(x,y)$ in the (x,y) plane. The region where $X > Y$ is shaded, corresponding to the area where $y < x$. The line $x=y$ is shown, and the region below it is shaded with vertical lines. The axes are labeled x and y .

Subsection 5

Conditional density

Definition of conditional density

Definition (Conditional density)

Let (X, Y) be random variables with joint density $f_{XY}(x, y)$. Let $f_X(x)$ and $f_Y(y)$ be the marginal densities.

- For a such that $f_X(a) > 0$ the conditional density of Y given $X = a$, denoted $f_{Y|X=a}(y)$, is defined as

$$\int_{-\infty}^{\infty} f_{Y|X=a}(y) dy = 1$$

$$f_{Y|X=a}(y) = \frac{f_{XY}(a, y)}{f_X(a)}$$

slice of joint at $x=a$
 $f_X(a) = \int_{-\infty}^{\infty} f_{XY}(a, y) dy$

- For b such that $f_Y(b) > 0$, the conditional density of X given $Y = b$, denoted $f_{X|Y=b}(x)$, is defined as

$$f_{X|Y=b}(x) = \frac{f_{XY}(x, b)}{f_Y(b)}$$

Properties of conditional density

- Both the conditional densities are valid densities in one dimension. So, the “conditional” random variables $(Y|X = a)$ and $(X|Y = b)$ are well-defined.
- Joint = Marginal ~~times~~^{*} Conditional, for $x = a$ and $y = b$ such that $f_X(a) > 0$ and $f_Y(b) > 0$

$$\underline{f_{XY}(a, b) = f_X(a)f_{Y|X=a}(b) = f_Y(b)f_{X|Y=b}(a)}$$

- The above is usually written as $f_{XY}(x, y) = f_X(x)f_{Y|X=x}(y) = f_Y(y)f_{X|Y=y}(x)$

Examples: Uniform

- Uniform on unit square
- $(X, Y) \sim \text{Uniform}(D)$, where
 $D = [0, 1/2] \times [0, 1/2] \cup [1/2, 1] \times [1/2, 1]$



$$Y|X=a \sim \text{Uniform}[0, 1] \\ 0 < a < 1$$

$$X|Y=b \sim \text{Uniform}[0, 1] \\ 0 < b < 1$$

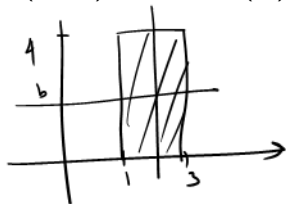


$$Y|X=a \sim \text{Uniform}[0, 1/2] \\ 0 < a < 1/2$$

$$Y|X=a \sim \text{Uniform}[1/2, 1] \\ 1/2 < a < 1$$

Examples: More uniform

- $(X, Y) \sim \text{Uniform}(D)$, where $D = [1, 3] \times [0, 4]$
- $(X, Y) \sim \text{Uniform}(D)$, where $D = [0, 1] \times [0, 1] \cup [1, 2] \times [0, 2]$
- $(X, Y) \sim \text{Uniform}(D)$, where $D = \{(x, y) : x + y < 2, x > 0, y > 0\}$

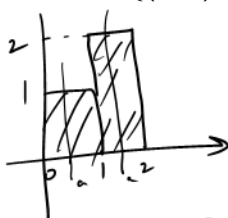


$$Y|X=a \sim \text{Uniform}[0, 4]$$

$$1 < a < 3$$

$$X|Y=b \sim \text{Uniform}[1, 3]$$

$$0 < b < 4$$

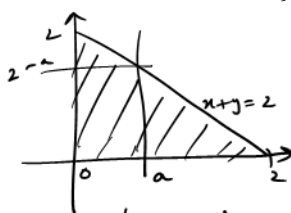


$$Y|X=a \sim \text{Uniform}[0, 1]$$

$$0 < a < 1$$

$$Y|X=a \sim \text{Uniform}[0, 2]$$

$$1 < a < 2$$



$$Y|X=a \sim \text{Uniform}[0, 2-a]$$

$$0 < a < 2$$

$$f_{Y|X=a}(y) = \begin{cases} \frac{1}{2-a}, & 0 < y < 2-a \\ 0, & \text{else.} \end{cases}$$

Problem

Consider the joint density

$$f_{XY}(x, y) = \begin{cases} x + y, & 0 < x, y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

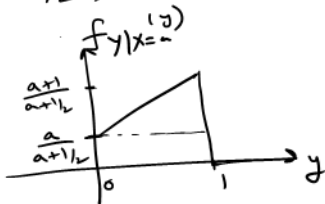
Find the conditionals.

$$f_X(x) = \int_0^1 (x+y) dy = \left(xy + \frac{y^2}{2} \right) \Big|_0^1 = x + \frac{1}{2}, \quad 0 < x < 1$$

$$f_Y(y) = y + \frac{1}{2}, \quad 0 < y < 1$$

$$f_{Y|X=a}(y) = \frac{a+y}{a+\frac{1}{2}}, \quad 0 < y < 1$$

$0 < a < 1$



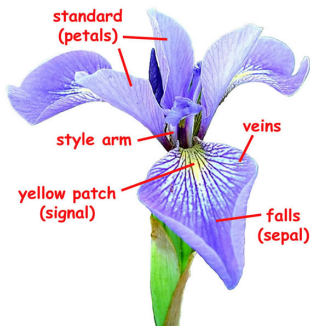
Subsection 6

From data to distribution

Iris data set

- Iris flower

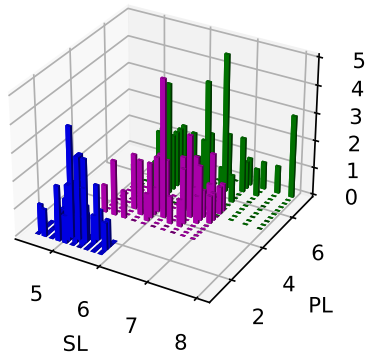
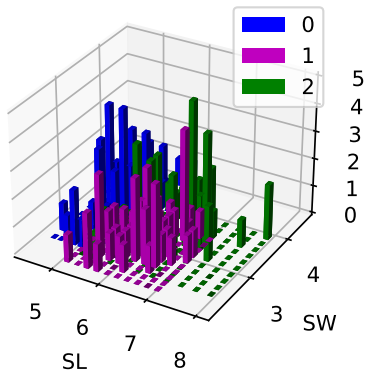
- ▶ 3 classes of irises: 0, 1 and 2
 - ★ 50 instances in each class
- ▶ Data (cm)
 - ★ sepal length (SL), sepal width (SW), petal length (PL), petal width (PW)
- ▶ Classification
 - ★ Given data, find class



(image source: fs.fed.us)

How to statistically describe (class, SL, SW, PL, PW)?

2D histograms: Class with (SL, SW) and (SL, PL)



- Notice how (SL, PL) seems to almost entirely separate the classes
- Notice the number of points per bin - very low

One discrete and two continuous random variables

How to describe joint distributions involving one discrete and two continuous variables?

One discrete and two continuous random variables

How to describe joint distributions involving one discrete and two continuous variables?

- (X, Y, Z) : X discrete and (Y, Z) continuous
- X has range T_X and PMF p_X
- For $x \in T_X$, “conditional” joint density $f_{YZ|X=x}(y, z)$
- Joint density $f_{YZ}(y, z) = \sum_{x \in T_X} p_X(x) f_{YZ|X=x}(y, z)$

Some issues in going from data to distribution

- When doing a histogram, there should be enough data points in each bin
- Typical situation: you will not have enough data to do stable histograms
 - ▶ Iris: only 50 instances per class
- In practice, the actual distribution is very difficult to know or estimate. However, it is good to have a sense of the distribution or the support at least.

Diabetes data set: 442 diabetes patients

- Ten baseline variables: age (yrs), sex (F/M), body mass index, average blood pressure, and six blood serum measurements
 - ▶ s1 tc, T-Cells (a type of white blood cells)
 - ▶ s2 ldl, low-density lipoproteins
 - ▶ s3 hdl, high-density lipoproteins
 - ▶ s4 tch, thyroid stimulating hormone
 - ▶ s5 ltg, lamotrigine
 - ▶ s6 glu, blood sugar level
- A quantitative measure of disease progression one year after baseline

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- A quantitative measure of disease progression one year after baseline

How would you start to describe the “distribution” from above data?

- Data is not sufficient to be confident about joint distribution of several variables
- Usual goal (called regression): predict disease progression as a function of baseline variables
- Typically, some assumptions are made on the distribution to help in prediction
 - ▶ It is good to justify these assumptions from the data